

APOLLO SOLAR VENTURES INC.

NUTECH Hardware & Software Solutions

NUTECH School System

cafeteria-pos.html

Staffed Cashier Terminal

Application Documentation

Prepared for Bernard P. Jocson · Developer: Carlos

Date: 22 June 2026

Contents

1. Overview

cafeteria-pos.html is the **staffed cashier terminal**. The cashier taps a card, sees the wallet balance, rings up menu items, charges the wallet, and prints a receipt. It also handles top-ups, linking new cards (enroll), and reassigning a card. It ships **live**.

2. Screens / tools

- **Checkout:** tap card → balance + menu → build cart → pay → receipt.
- **Top-up:** add money to a wallet.
- **Enroll:** link a blank card to an account.
- **Reassign:** move a card to a new owner (Admin PIN gated).
- **Settings:** webhook URL, RFID reader (HID / Arduino), demo flag.

3. Configuration

```
DEMO_MODE: false           // POS ships live
N8N_BASE: ".../webhook/cafeteria"
RFID_SOURCE: "hid"         // or "arduino"
ARDUINO_BAUD: 115200
REQUIRE_FP: false         // fingerprint removed
ADMIN_PIN: "1234"          // reassign tool
```

4. Backend actions used

Action	Sends / does
verify	card tap → identity + balance
purchase	charge wallet; writes sale + line items + stock
topup	add money to wallet
enroll	link a blank card to an account
menu	load menu items
account-info	look up a wallet
reassign	move card to a new owner

5. Hardware

- **RFID:** Arduino + PN532 over USB (Web Serial). **Confirmed working.**
- **Fingerprint:** removed (DigitalPersona offline) — card-only checkout.
- **Printer (optional):** XP-58IIH via print-bridge on port 9002.

6. Running & go-live

1. **Serve over localhost/HTTPS** for the Arduino reader.

2. Already live — just confirm the webhook in Settings and that n8n is **Active**.

7. Notes

- **One reader per page** — the POS owns the cashier-PC reader; don't open the gate/portal on the same machine simultaneously.
- **Card-only now** — purchases never wait on a fingerprint.